

III. Heart Disease Dataset Analysis Using Mosaic Plots and Cotabplots

```
library(vcd)

heart_data <- read.csv("heart-1.csv")

heart_data$HeartDisease <- factor(heart_data$HeartDisease, levels =
c(0, 1), labels = c("No", "Yes"))
heart_data$ST_Slope <- factor(heart_data$ST_Slope)
heart_data$ExerciseAngina <- factor(heart_data$ExerciseAngina)
heart_data$RestingECG <- factor(heart_data$RestingECG)

head(heart_data)
```

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	
MaxHR								
1	40	M	ATA	140	289	0	Normal	172
2	49	F	NAP	160	180	0	Normal	156
3	37	M	ATA	130	283	0	ST	98
4	48	F	ASY	138	214	0	Normal	108
5	54	M	NAP	150	195	0	Normal	122
6	39	M	NAP	120	339	0	Normal	170

	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
1	N	0.0	Up	No
2	N	1.0	Flat	Yes
3	N	0.0	Up	No
4	Y	1.5	Flat	Yes
5	N	0.0	Up	No
6	N	0.0	Up	No

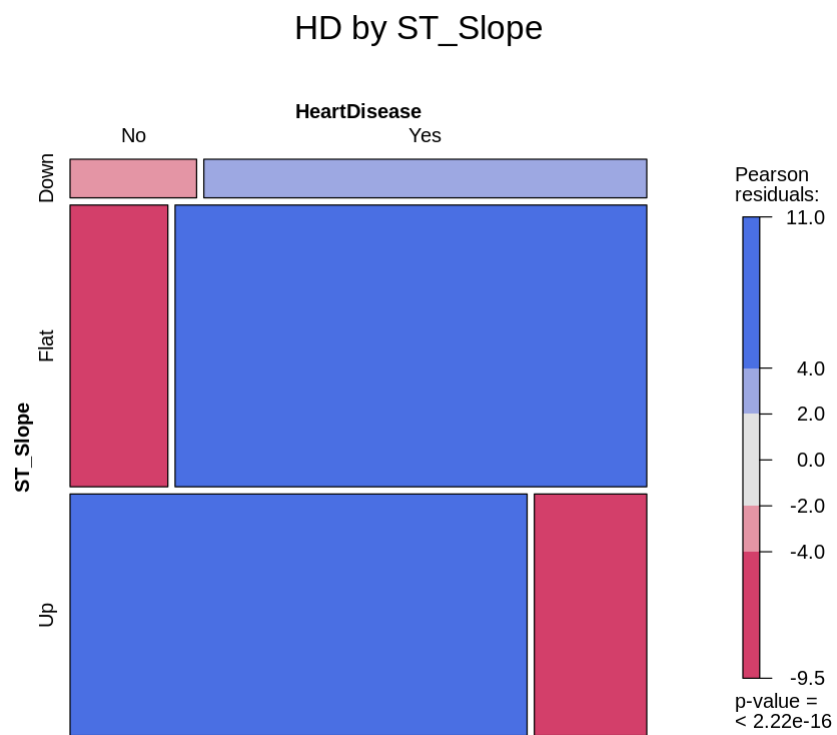
1. Mosaic Plots: Predictors vs. HeartDisease

```
par(mfrow = c(1, 3))

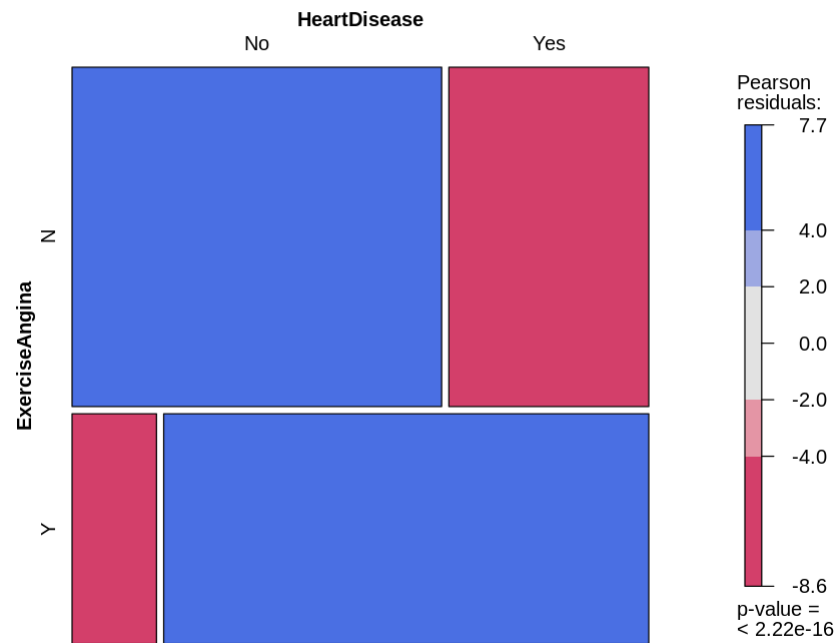
mosaic(~ ST_Slope + HeartDisease, data = heart_data,
      main = "HD by ST_Slope", shade = TRUE, legend = TRUE)

mosaic(~ ExerciseAngina + HeartDisease, data = heart_data,
      main = "HD by Exercise Angina", shade = TRUE, legend = TRUE)
```

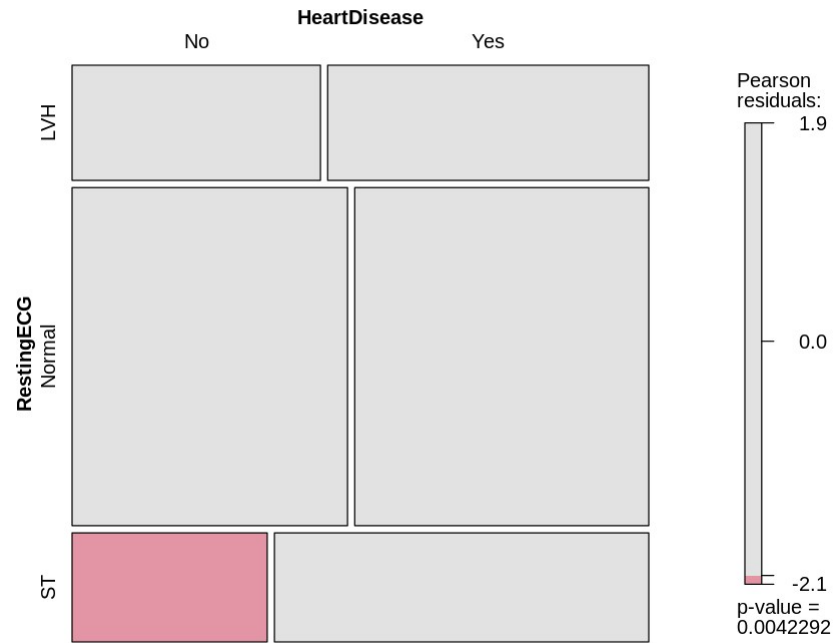
```
mosaic(~ RestingECG + HeartDisease, data = heart_data,  
       main = "HD by Resting ECG", shade = TRUE, legend = TRUE)
```



HD by Exercise Angina



HD by Resting ECG



1. Which variables appear useful for predicting HeartDisease?
 - The ST_Slope appears to be useful in predicting HeartDisease. With St_Slope those with an Up slope are more likely to not get HeartDisease and those who have a Flat or Down slope are likely to get a HeartDisease.
 - With ExerciseAngina also appearing to be useful in predicting HeartDisease. With ExerciseAngina those who do experience chest pain during exercise are more likely to get a HeartDisease compared to those who don't.
2. Which variables appear less useful?
 - The RestingECG showcases that the proportions of heart disease accross all three types of abnormality are all relatively similar. Making it not useful in predicting HeartDisease.

2. Relationships Among Predictors

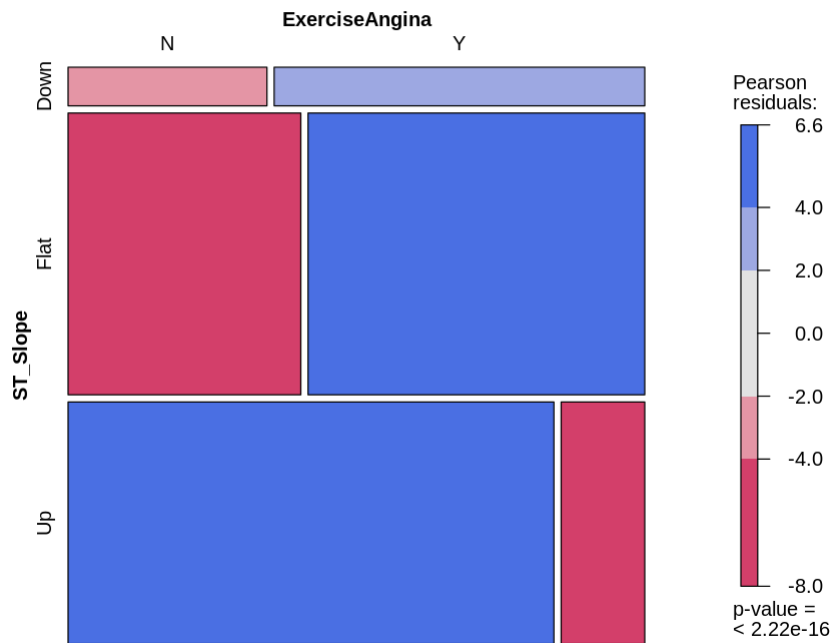
```
par(mfrow = c(1, 3))

mosaic(~ ST_Slope + ExerciseAngina, data = heart_data,
      main = "ST_Slope vs ExerciseAngina", shade = TRUE)

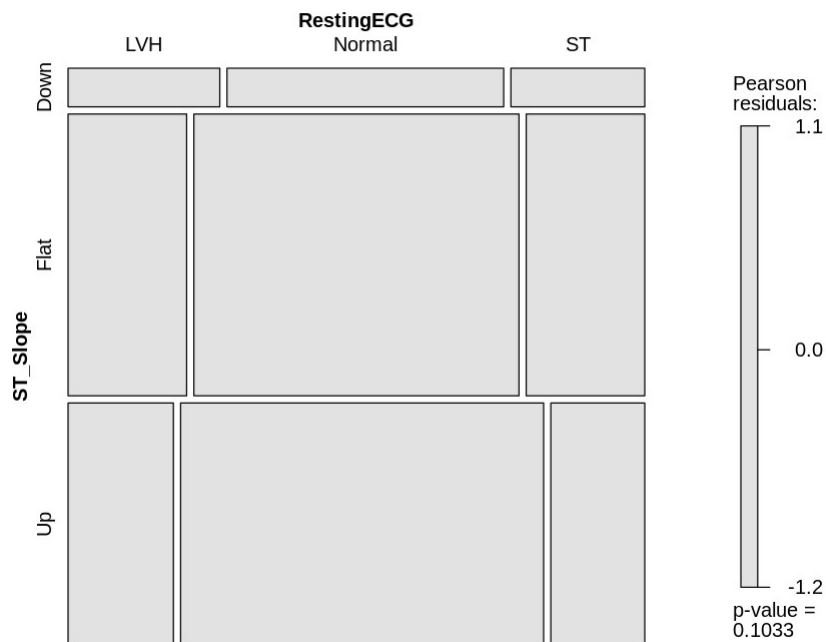
mosaic(~ ST_Slope + RestingECG, data = heart_data,
      main = "ST_Slope vs RestingECG", shade = TRUE)

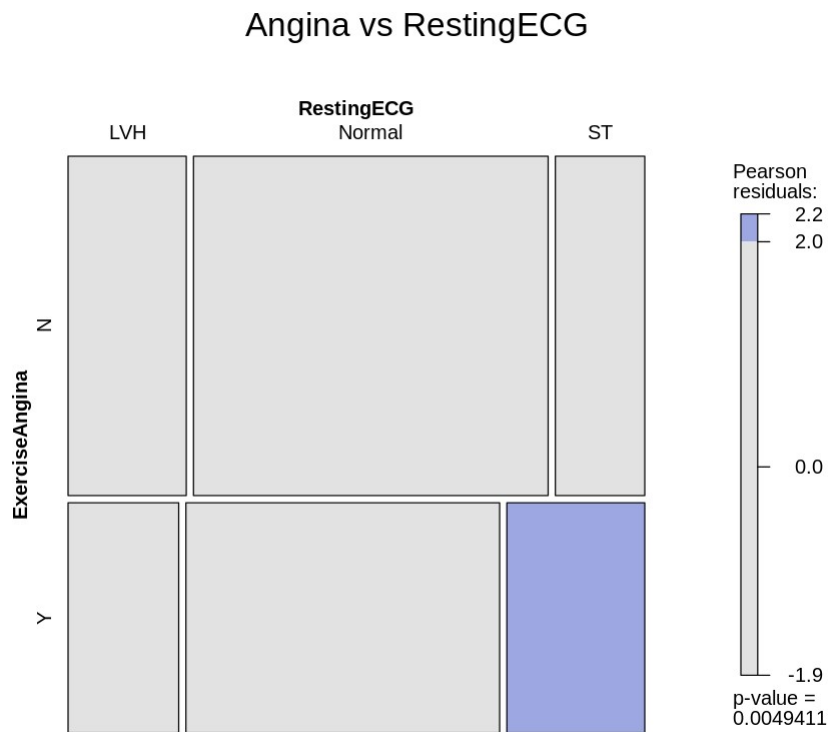
mosaic(~ ExerciseAngina + RestingECG, data = heart_data,
      main = "Angina vs RestingECG", shade = TRUE)
```

ST_Slope vs ExerciseAngina



ST_Slope vs RestingECG



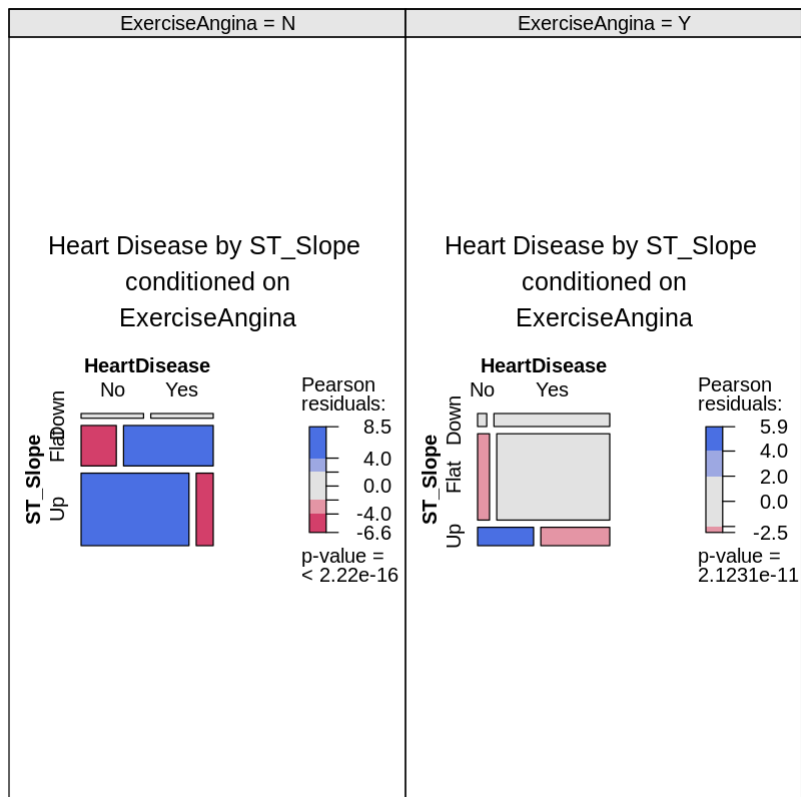


Discussion:

Both St_Slope and ExerciseAngina are likely to be redundant as both of these variables are an indicator of a person's heart health. As a person who does feel chest pain when exercising is often likely to also have a Flat or Down Slope which both mean the same thing that the person's heart is not in good shape.

3. Cotabplot Investigation

```
cotabplot(~ ST_Slope + HeartDisease | ExerciseAngina, data =
heart_data,
  panel = cotab_mosaic, shade = TRUE,
  main = "Heart Disease by ST_Slope \n conditioned on \nExerciseAngina",
  main_gp = gpar(fontsize = 15))
```



Intepretation:

According to the cotabplot the combination of St_Slope and ExerciseAngina does in fact provide more predictive than individually.

It can be seen that when a person does not feel chest pain during exercise and has an Up slope they are almost guranteed to not have heart disease.

And the opposite is true with a person who does feel chest pain during exercise and has a Flat or Down Slope is almost guranteed to have heart disease.